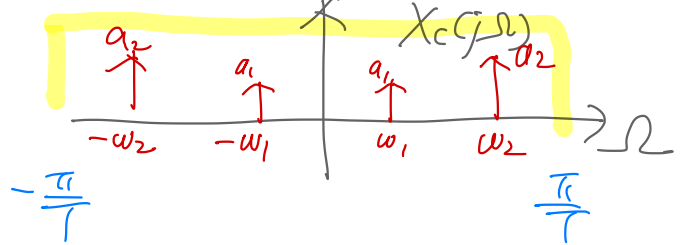
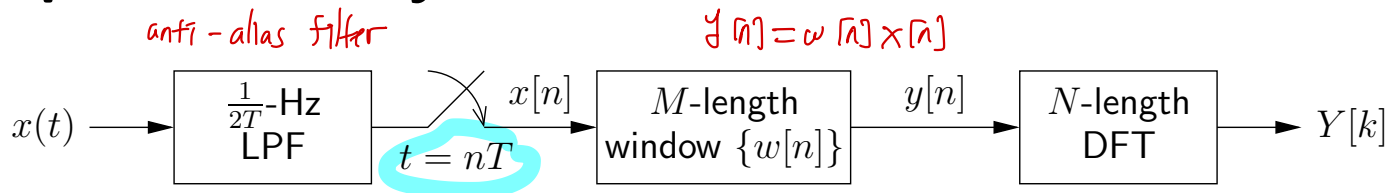


$$x(t) = a_1 \cos(\omega_1 t) + a_2 \cos(\omega_2 t)$$



Spectral Analysis:

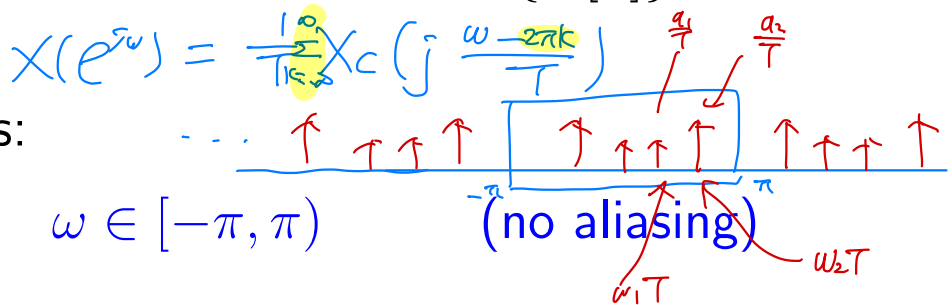


Goal: Characterize the sinusoidal components in $x(t)$.

Method: Sample, window (i.e., $y[n] = w[n]x[n]$), and transform non-zero samples to the DFT domain.

Questions: How to choose M , $\{w[n]\}$, and N ?
How to interpret DFT outputs $\{Y[k]\}$?

Recall key relationships:



$$X(e^{j\omega}) = \frac{1}{T} X_c(j\frac{\omega}{T}), \quad \omega \in [-\pi, \pi)$$

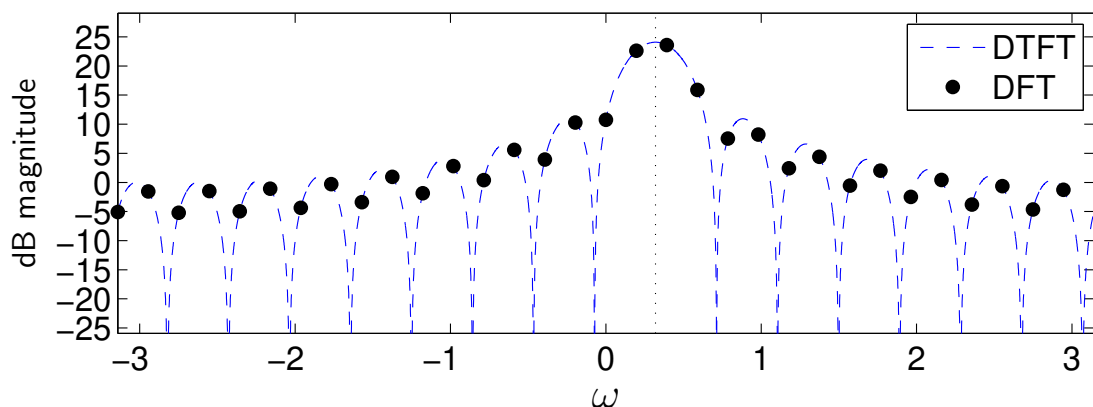
$$Y(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \quad (y[n] = w[n]x[n])$$

$$Y[k] = Y(e^{j\frac{2\pi}{N}k}) \quad (\text{if } N \geq M)$$

$$x(t) = e^{j\omega_0 t}$$

complex sinusoid at $\omega_0 = 0.32$, $M=16$, $N=32$

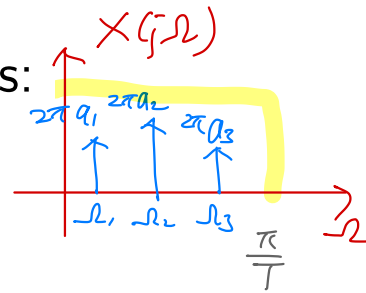
Example:



Spectral Analysis (cont.):

Suppose our signal is a sum of L complex sinusoids:

$$x(t) = \sum_{l=1}^L a_l e^{j\Omega_l t} \quad \text{for } |\Omega_l| < \frac{\pi}{T}.$$



Then, using the relationships from the previous page,

$$X_c(j\Omega) = 2\pi \sum_{l=1}^L a_l \delta(\Omega - \Omega_l)$$

$$X(e^{j\omega}) = \frac{2\pi}{T} \sum_{l=1}^L a_l \delta\left(\frac{\omega}{T} - \Omega_l\right) \quad \text{for } \omega \in [-\pi, \pi)$$

$$\begin{aligned} Y(e^{j\omega}) &= \frac{1}{T} \int_{-\pi}^{\pi} \sum_{l=1}^L a_l \delta\left(\frac{\xi}{T} - \Omega_l\right) W(e^{j(\omega-\xi)}) d\xi \\ &= \sum_{l=1}^L a_l \int_{-\pi/T}^{\pi/T} \delta(\theta - \Omega_l) W(e^{j(\omega-\theta T)}) d\theta \quad \text{via } \theta = \frac{\xi}{T} \\ &= \sum_{l=1}^L a_l W(e^{j(\omega-\Omega_l T)}) \end{aligned}$$

... a sum of scaled and shifted window-DTFTs! Finally,

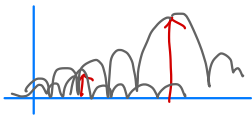
$$Y[k] = \sum_{l=1}^L a_l W\left(e^{j\left(\frac{2\pi}{N}k - \Omega_l T\right)}\right)$$

So we expect a DFT peak near $k \approx \frac{\Omega_l T N}{2\pi}$ for each $l = 1 \dots L$.

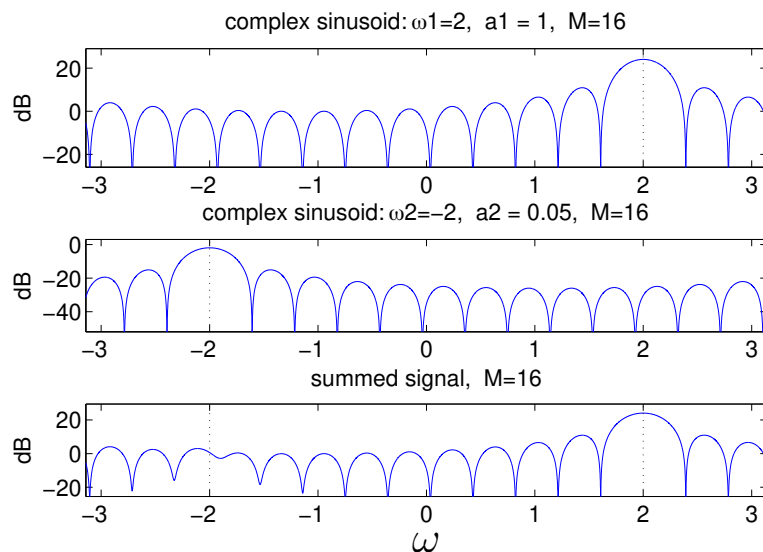
Effects of finite data-length:

With $L > 1$ complex sinusoids, the DFT peaks may not be where we expect! There are two main reasons for this:

1. Leakage: The energy at $\omega_l = \Omega_l T$ gets spread over all $\omega \in [-\pi, \pi]$ through the window's sidelobes.



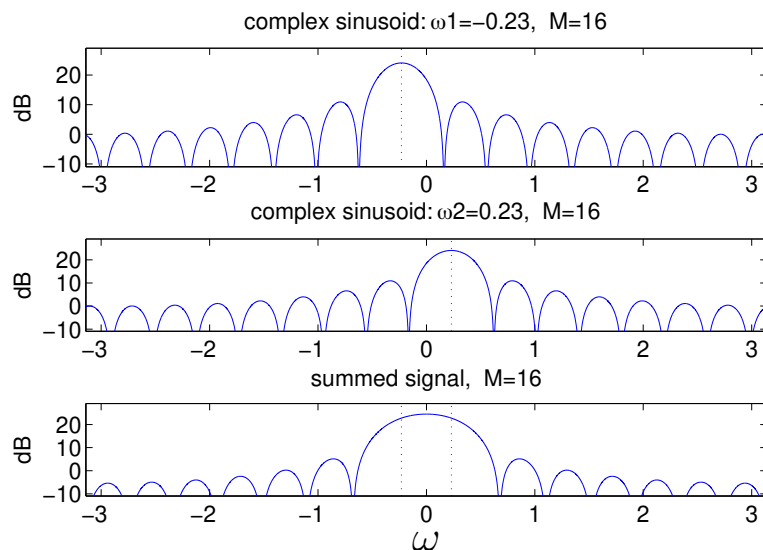
*Weak sinusoids
get buried!*



2. Limited Resolution: The energy at $\omega_l = \Omega_l T$ gets spread broadly over the window's main-lobe.

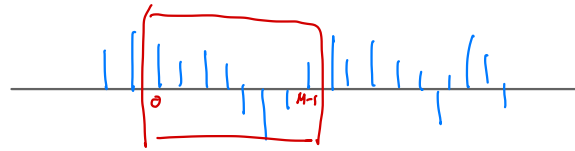


*Closely spaced
sinusoids
become indis-
tinguishable!*



Can **trade** between leakage & resolution via choice of window.

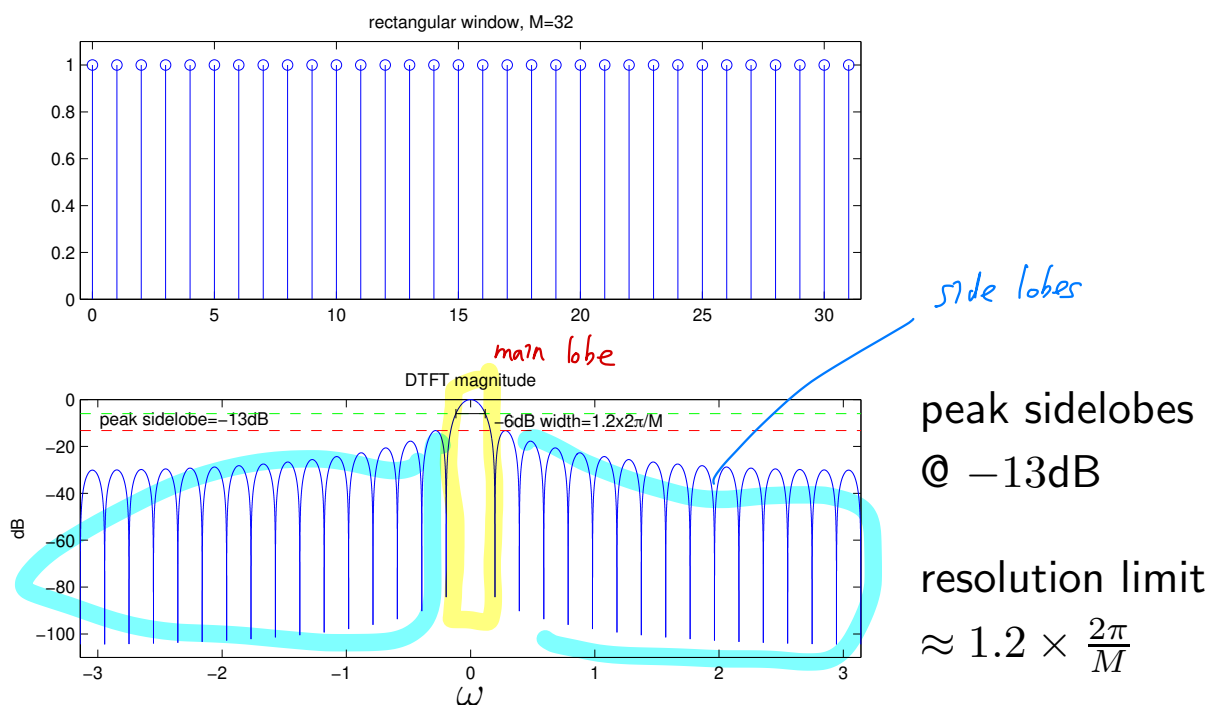
Rectangular window:



Recall that, for an M -length rectangular window, we saw

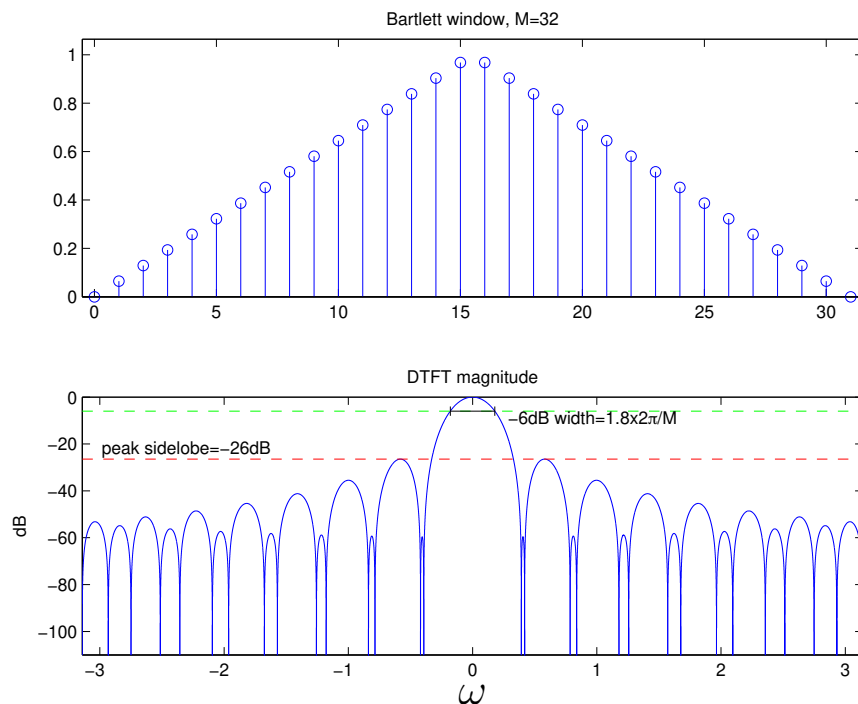
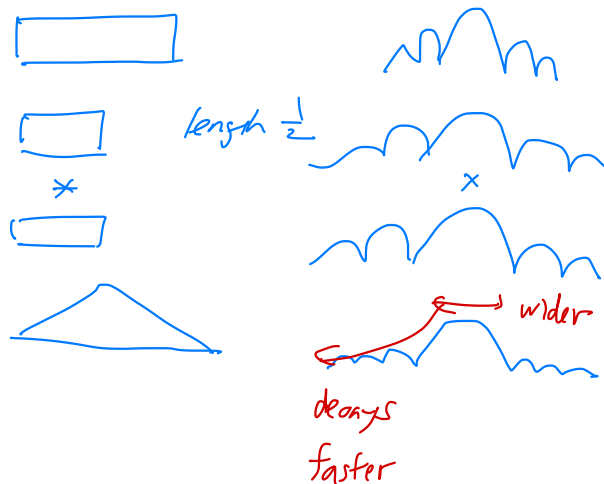
$$W(e^{j\omega}) = \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j\omega(M-1)/2}.$$

So, in time- and frequency-domain, this window looks like:



where “resolution limit” is “-6 dB bandwidth” and approximates the minimum spacing between recognizable sinusoids (of equal amplitude).

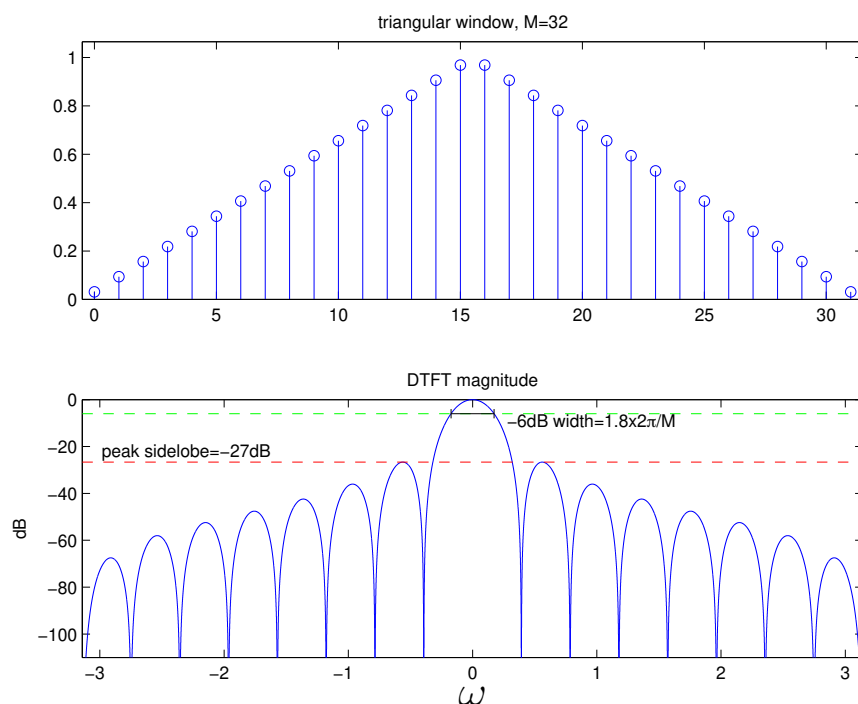
Bartlett window:



peak sidelobes
@ -26dB

resolution limit
 $\approx 1.8 \times \frac{2\pi}{M}$

Triangular window:

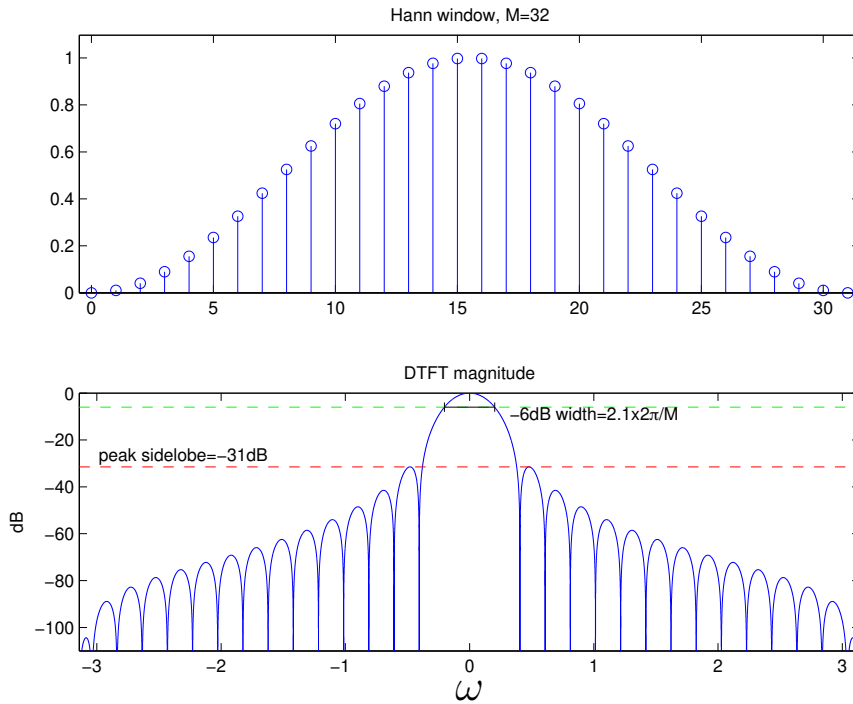


improved: edge
points are not
zero-valued!

peak sidelobes
@ -27dB

resolution limit
 $\approx 1.8 \times \frac{2\pi}{M}$

Hann window:

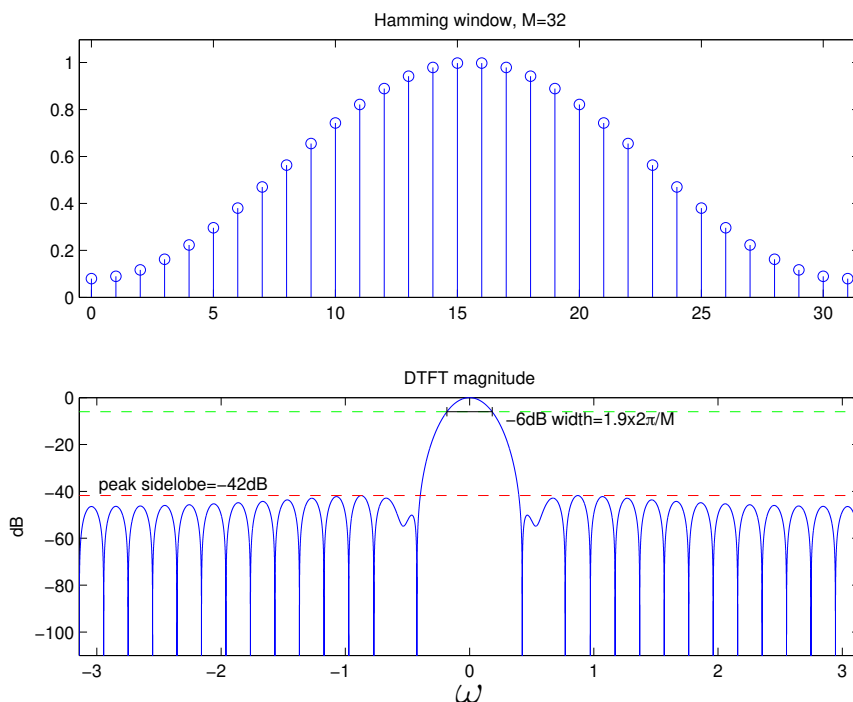


$$w[n] = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi}{M}n\right)$$

peak sidelobes
@ -31dB

resolution limit
 $\approx 2.1 \times \frac{2\pi}{M}$

Hamming window:

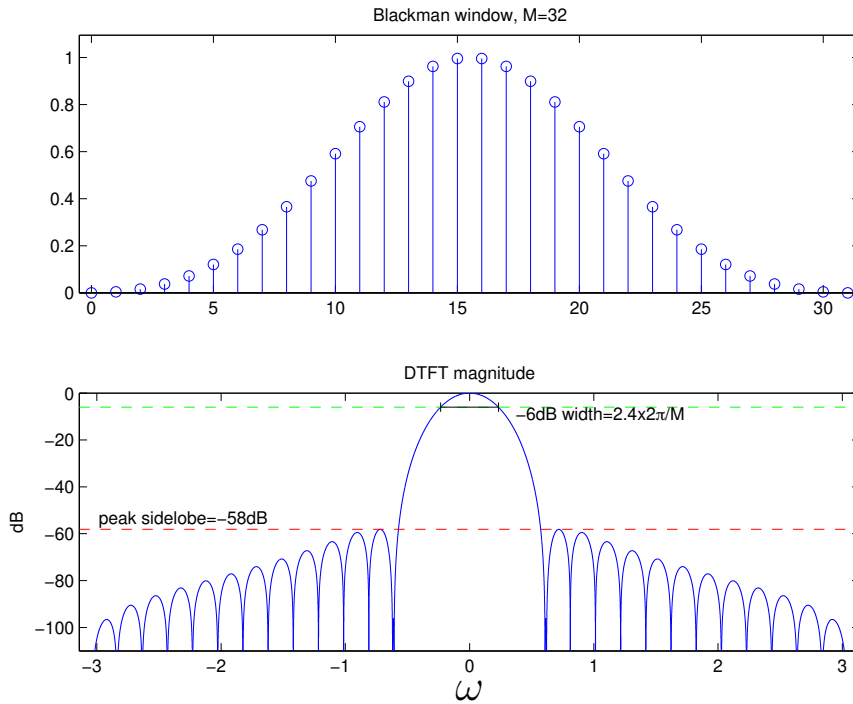


$$w[n] = 0.538 - 0.462 \cos\left(\frac{2\pi}{M}n\right)$$

peak sidelobes
@ -42dB

resolution limit
 $\approx 1.9 \times \frac{2\pi}{M}$

Blackman window:

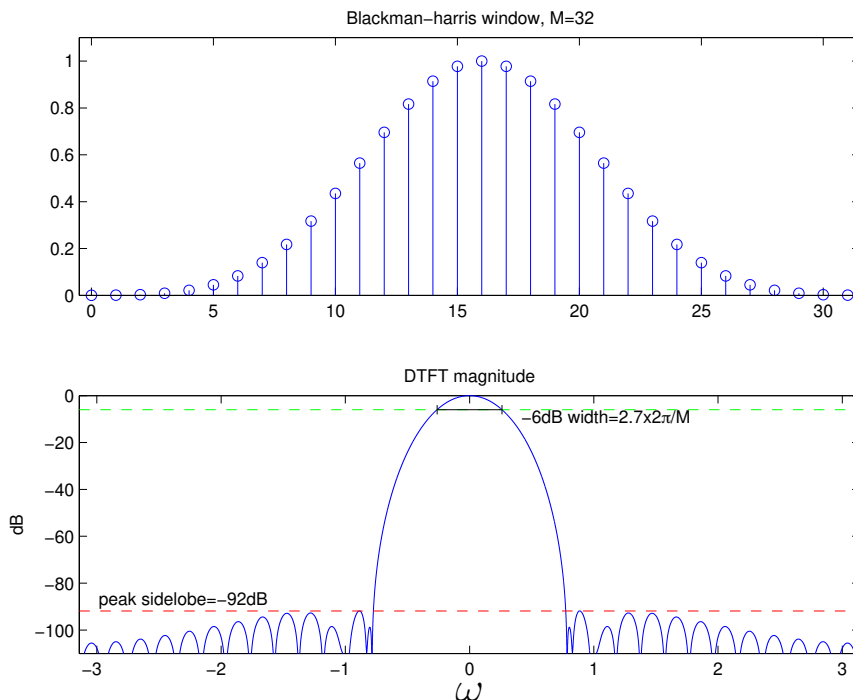


$$w[n] = 0.42 - \frac{1}{2} \cos\left(\frac{2\pi}{M}n\right) + 0.08 \cos\left(\frac{4\pi}{M}n\right)$$

peak sidelobes
@ -58dB

resolution limit
 $\approx 2.4 \times \frac{2\pi}{M}$

Blackman-harris window:



optimized 4-term

peak sidelobes
@ -92dB

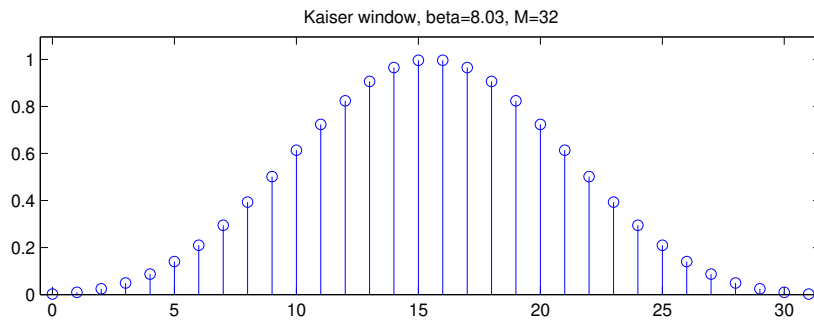
resolution limit
 $\approx 2.7 \times \frac{2\pi}{M}$

Kaiser window:

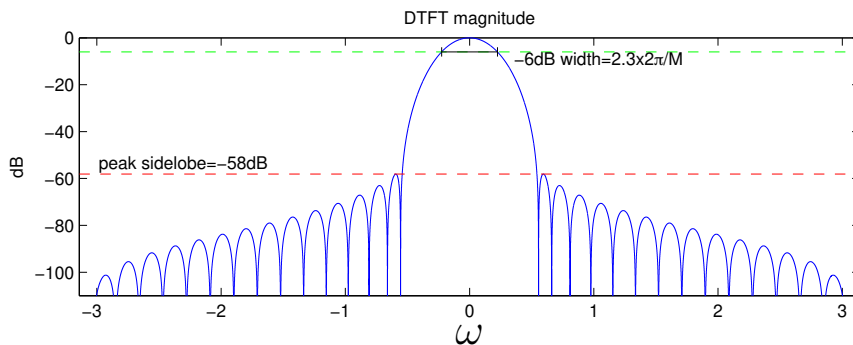
Has adjustable parameter β . For peak sidelobe level $-S_{\text{dB}}$,

$$\beta = \begin{cases} 0 & S_{\text{dB}} = 13.26 \\ 0.76609(S_{\text{dB}} - 13.26)^{0.4} + 0.09834(S_{\text{dB}} - 13.26) & 13.26 < S_{\text{dB}} \leq 60 \\ 0.12438(S_{\text{dB}} + 6.3) & 60 < S_{\text{dB}} \leq 120 \end{cases}$$

For example, $\beta = 8.03$ gives $S_{\text{dB}} = 59$ and, when $M = 32$,



based on modified
Bessel function



peak sidelobes
@ -58dB

resolution limit
 $\approx 2.3 \times \frac{2\pi}{M}$

which is very similar (and slightly better) than Blackman.

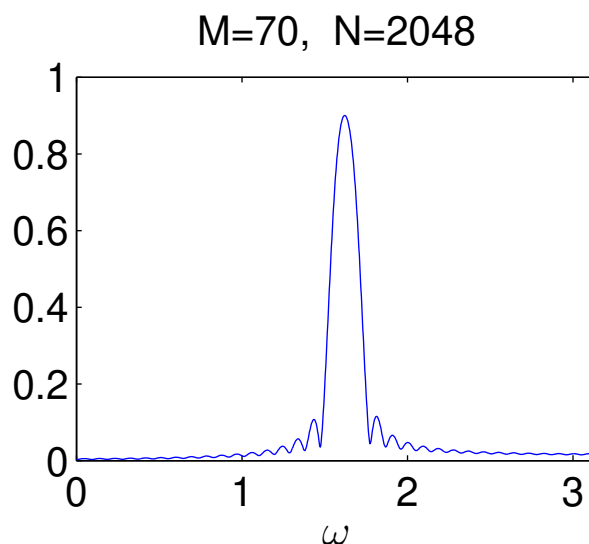
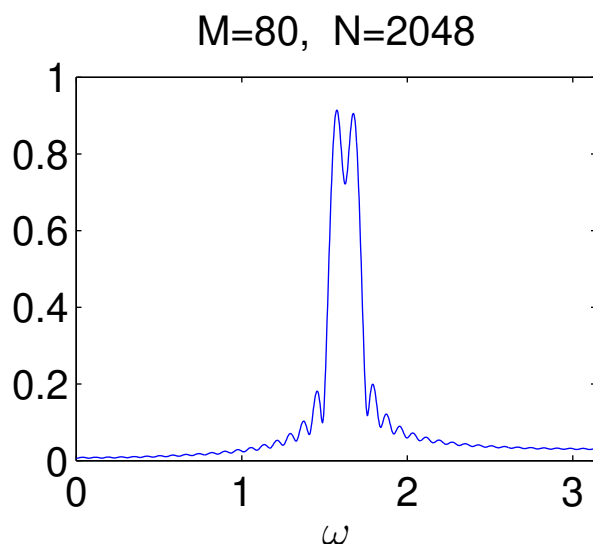
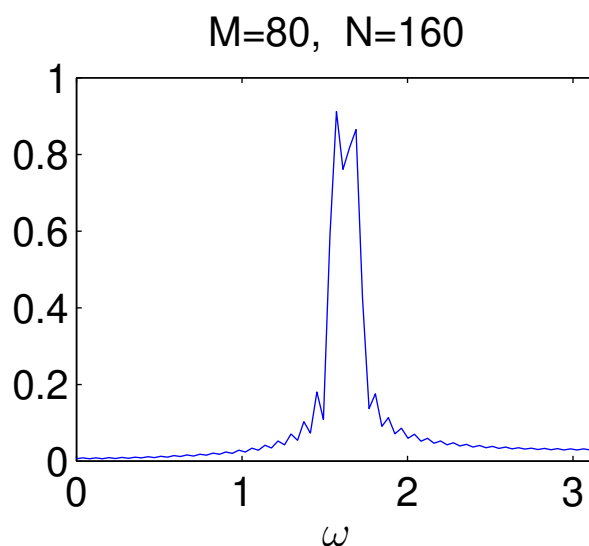
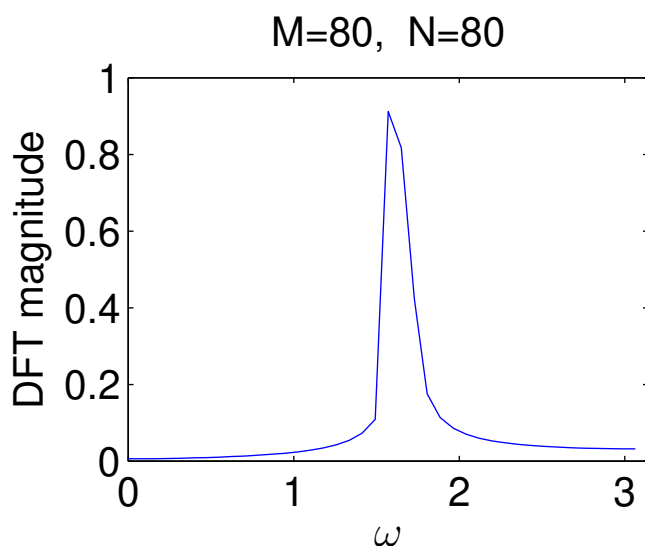
The approximate tradeoff between peak sidelobe level $-S_{\text{dB}}$ and -6dB mainlobe width Δ_{ω} (in rad/sample) is given by

$$\Delta_{\omega} \approx 0.029(29 + S_{\text{dB}} - 0.003 S_{\text{dB}}^2) \frac{2\pi}{M}.$$

Spectral analysis example:



- Real sines at 30 Hz & 32 Hz, $\frac{1}{T} = 120 \text{ Hz}$. ($\Delta\omega = \frac{2 \times 2\pi}{120}$)
- Recall: M -length data, N -length DFT.
- Rectangular window: resolution limit $\frac{1.2 \times 2\pi}{M}$ ($\Rightarrow M > 72$).



- In general:

Need DFT length $N \geq 2M$, but prefer $N \gg 2M$.

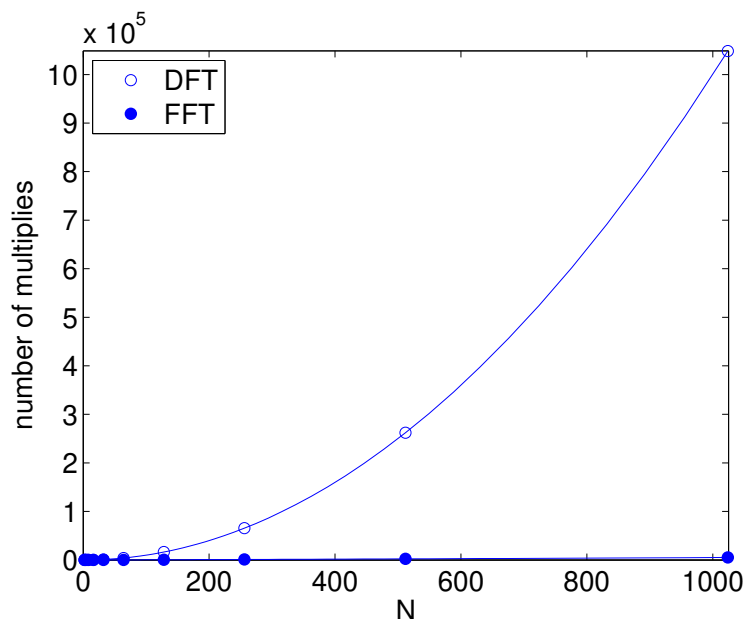
The fast Fourier transform (FFT):

- A family of algorithms to quickly compute the DFT.
- Recall that the N -DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn}$$

consumes N (complex-valued) multiplications per output, and thus N^2 multiplications total.

- We will describe the so-called “radix-2 decimate-in-time” FFT, which works when N is power-of-two, and consumes approximately $\frac{N}{2} \log_2 N$ multiplications.



For large N , the FFT is much cheaper than the brute-force DFT!

The radix-2 FFT:

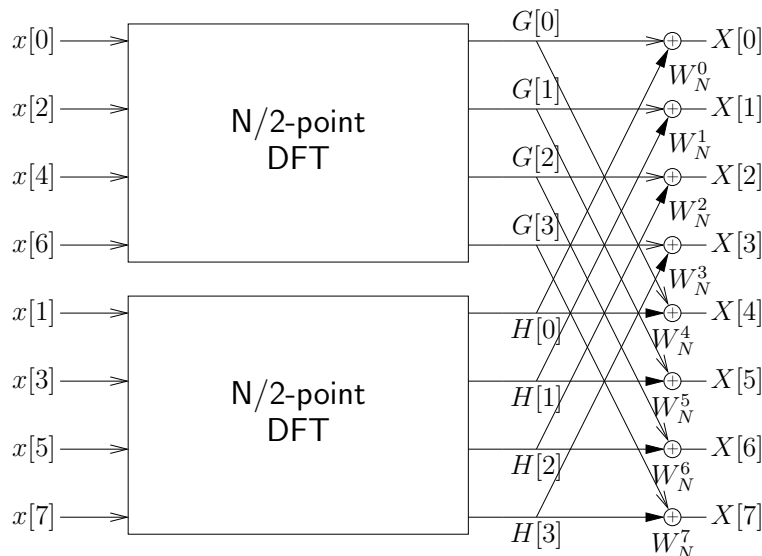
Notice that

$$\begin{aligned}
 X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad \text{for } W_N^q \triangleq e^{-j\frac{2\pi}{N}q} \\
 &= \sum_{n \text{ even}} x[n] W_N^{kn} + \sum_{n \text{ odd}} x[n] W_N^{kn} \\
 &= \sum_{m=0}^{N/2-1} x[2m] W_N^{k2m} + \sum_{m=0}^{N/2-1} x[2m+1] W_N^{k(2m+1)}
 \end{aligned}$$

Because $W_N^{2q} = W_{N/2}^q$ and $W_N^{q+p} = W_N^q W_N^p$, we can write

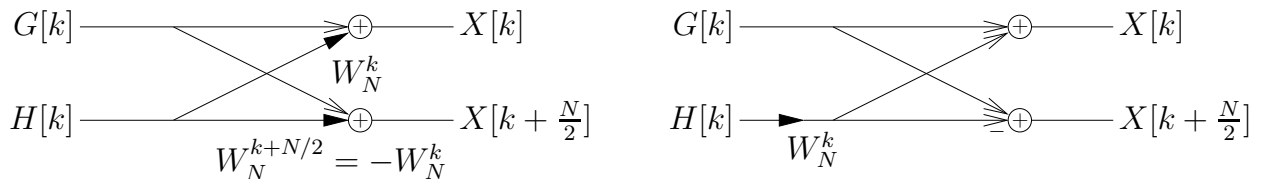
$$\begin{aligned}
 X[k] &= \underbrace{\sum_{m=0}^{N/2-1} x[2m] W_{N/2}^{km}}_{\triangleq G[k]} + W_N^k \underbrace{\sum_{m=0}^{N/2-1} x[2m+1] W_{N/2}^{km}}_{\triangleq H[k]}
 \end{aligned}$$

Note: $G[k]$ is an $\frac{N}{2}$ -DFT of the even samples, and $H[k]$ is an $\frac{N}{2}$ -DFT of the odd samples. Both are $\frac{N}{2}$ -periodic in k .

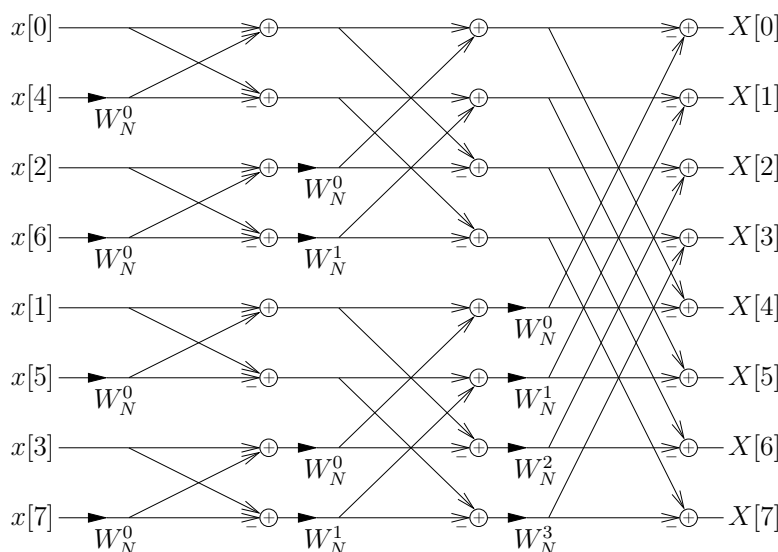


The radix-2 FFT (cont.):

- The final processing stage can be simplified from N mults to $\frac{N}{2}$ mults because the following are equivalent:



- Let's examine the complexity of this split-in-half method:
 - direct N -DFT: N^2 mults,
 - split-in-half: $(\frac{N}{2})^2 + (\frac{N}{2})^2 + \frac{N}{2} = \frac{1}{2}(N^2 + N)$ mults,
 Since $\frac{1}{2}(N^2 + N) < N^2$ for any $N > 1$, *it pays to split!*
- Thus, we split each $\frac{N}{2}$ -DFT into two $\frac{N}{4}$ -DFTs (combining their outputs using twiddle factors), repeat with each $\frac{N}{4}$ -DFT, and so on, finally obtaining



In total, $(\log_2 N)$ stages $\times \frac{N}{2} \frac{\text{mults}}{\text{stage}} = \frac{N}{2} \log_2 N$ mults.